

Prime-factor density reversals and the spectrum of bounded prime gaps

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Abstract

We study strict descents followed by later strict ascents in the density of the k th smallest distinct prime factor. For a fixed local probability law satisfying $\eta_p^{-1} = p/\nu + \beta + o(1)$, with $\nu > 0$, every rank has a finite, attained maximal number $N_\eta(k)$ of separated reversals. Its double-logarithmic upper growth rate is exactly $\sup_{H \geq 2} \gamma_H/(H + \nu)$, where γ_H measures the actual frequency of consecutive prime gaps at most H . The positive rate has a finite phase profile and applies to limiting factorization densities of products of distinct affine forms in the sum of two primes. For the ordinary and generic odd laws we also obtain double-exponential bounds at every sufficiently large rank and an exact equivalence between recurring short gaps and unbounded late-ascent supply. The proofs combine uniform symmetric-polynomial estimates with a finite selection argument using periodic composite blocks. The maintained results are formalized in Lean. The transfers do not establish a new numerical prime-gap bound.

1 Introduction and historical context

Let $d_k(p)$ be the natural density of positive integers whose k th smallest *distinct* prime factor is p . The question whether $p \mapsto d_k(p)$ first rises and then falls is now called Erdős Problem 690. Our subject is the finer structure left after unimodality fails: the number of separated reversals, their possible locations, and the arithmetic information encoded by their growth.

Erdős formulated the question in 1979 [1, p. 75]. The same paper discusses the typical scale $\log \log p_k(n) \sim k$ for prime factors of integers. This concerns where probability mass lies, not the mode of the individual-prime density. Erdős–Tenenbaum [2, Corollary 5] later corrected the earlier assertion about the mode: a maximizing prime satisfies $\log p \sim k/\log k$, rather

than $\log p \sim k$. Neither statement counts reversals. The occurrence of a double exponential in classical prime-factor distribution does not by itself supply the theorem below.

Cambie [3, Theorem 5] proved unimodality for $k = 1, 2, 3$ and verified its failure for $4 \leq k \leq 20$. His Theorem 3 gives superpolynomially many local maxima for the different density $\delta_1(n, m)$ in Erdős Problem 692: the density of integers having exactly one divisor in $\{n + 1, \dots, m - 1\}$. Its moving variable is m , with n fixed. It is not a reversal count for $d_k(p)$.

Wang–Crapis [4, Theorem 1.1] proved non-unimodality for every $k \geq 4$. Their proof compares a symmetric-polynomial ratio with prime gaps and produces a descent followed by an ascent. We extend the comparison to local laws and make the construction quantitative. The question here is how the maximal number of witnesses grows with k . Its exact rate also requires a converse: a sufficiently late ascent must come from an appropriately short gap.

The contribution is the resulting frequency spectrum, its realization in prime-pair arithmetic, and its late-ascent interpretation. Quantitative bounded-gap results are existing analytic inputs, not new sieve results of this paper. The public repository contains maintained proofs and a provenance ledger. This is an AI-assisted research draft; independent expert review and assessment of novelty remain outstanding.

2 The argument in plain English

Choose a position in an integer’s ordered list of prime factors, say the tenth distinct prime factor. For each prime, ask what fraction of sampled integers put that prime in the chosen position. These fractions form a sequence. A reversal means that the sequence drops and later rises again. We count occurrences separated far enough that they do not reuse the same stretch of the sequence.

Two effects compete. Moving to a larger prime lowers its own chance of dividing the integer. At the same time, the additional smaller primes make it easier to have the required number of preceding factors. An exact formula measures this competition. On the relevant scale, a short gap between consecutive primes favors a rise, whereas a long gap favors a drop.

The useful measuring scale is the logarithm of the logarithm of the prime. On that scale the available rise threshold is approximately the rank divided by the scale. This explains the double-exponential locations in the argument. Eventually even the shortest possible prime gap is too large for a rise, so

every fixed rank still has a final decreasing tail.

Many short gaps alone are not enough to count reversals. We must put a long gap before each selected short gap and keep the witnesses distinct. Periodic blocks of composite integers provide barriers that consecutive primes must jump. Grouping short gaps between the barriers and taking alternate occupied groups produces separated witnesses, losing only a fixed factor in their number.

Finally, different short-gap sizes can have different frequencies. Each size contributes its frequency divided by a threshold cost. The greatest contribution is exactly the reversal growth rate. For products of affine prime sums, the number of distinct factors changes that cost, while finitely many exceptional primes change only the initial details. The last sections explain the prime-gap information this recovers and the limits of that conclusion.

3 Definitions and main results

All logarithms are natural. Index the primes by $p_0 = 2, p_1 = 3, \dots$, and write $\mathcal{L}(x) = \log \log x$ for $x > 1$ and $\mathcal{E}(t) = \exp(\exp t)$. Ranks k are positive integers. A fixed *local law* consists of probabilities $0 \leq \eta_p \leq 1$. Define the finite products

$$M_r(p) = [z^r] \prod_{q < p} (1 - \eta_q + \eta_q z), \quad D_{\eta,k}(p) = \eta_p M_{k-1}(p), \quad (1)$$

with $M_r = 0$ for $r < 0$. For the ordinary law $\eta_p = 1/p$, the Chinese remainder theorem identifies $D_{\eta,k}$ with d_k . The second canonical law is

$$\eta_2^{\text{odd}} = 0, \quad \eta_p^{\text{odd}} = \frac{p-2}{(p-1)^2} \quad (p > 2). \quad (2)$$

Definition 3.1. A reversal at rank k consists of indices $d < a$ with

$$D_{\eta,k}(p_{d+1}) < D_{\eta,k}(p_d), \quad D_{\eta,k}(p_a) < D_{\eta,k}(p_{a+1}).$$

A family of m such witnesses is separated if $a_i + 1 < d_j$ for $i < j$. Its maximal attainable size, when finite, is $N_\eta(k)$.

This counts raw-density descent-ascent witnesses, not oscillations of a centered error or a smoothed approximation.

For integers $H \geq 0$, set

$$\begin{aligned} G_H(X) &= \#\{i : p_i \leq X, p_{i+1} - p_i \leq H\}, \\ \gamma_H &= \limsup_{\substack{X \rightarrow \infty \\ X \in \mathbb{N}}} \frac{\log \log(3 + G_H(X))}{\mathcal{L}(X)}, \\ \Lambda(\nu) &= \sup_{H \geq 2} \frac{\gamma_H}{H + \nu} \quad (\nu > 0). \end{aligned} \tag{3}$$

The 3 keeps the logarithms defined when counts vanish. We have $0 \leq \gamma_H \leq 1$ and $\gamma_0 = \gamma_1 = 0$.

Theorem 3.2 (Exact reversal spectrum). *Suppose $\nu > 0$, $\beta \in \mathbb{R}$, and, along sufficiently large primes,*

$$0 < \eta_p < 1, \quad \eta_p^{-1} = p/\nu + \beta + \rho(p), \quad \rho(p) \longrightarrow 0. \tag{4}$$

Every fixed-rank sequence $D_{\eta,k}$ is eventually nonincreasing, its reversal maximum is finite and attained, and

$$\limsup_{k \rightarrow \infty} \frac{\log \log(3 + N_\eta(k))}{k} = \Lambda(\nu) > 0. \tag{5}$$

The law, its parameters and its exceptional coordinates are fixed before k tends to infinity.

The weaker estimate $\eta_p = \nu/p + O(p^{-2})$ alone does not supply the vanishing difference of reciprocal errors used in this theorem.

Theorem 3.3 (Canonical bounds). *For either canonical law and every $\varepsilon > 0$, for all sufficiently large k ,*

$$\mathcal{E}(k/602) \leq N_\eta(k) \leq \mathcal{E}((1/3 + \varepsilon)k).$$

Each fixed-rank canonical sequence is eventually strictly decreasing.

The positive constant uses a conservative quantitative width-600 input; it is not a claim about the best known numerical prime-gap bound.

Theorem 3.4 (Affine arithmetic realization). *Fix $m \geq 1$, positive integers c_j , and integers d_j , with $c_i d_j \neq c_j d_i$ for $i \neq j$. Put*

$$F(p, q) = \prod_{j=1}^m (c_j(p + q) + d_j).$$

Sample all ordered prime pairs $p, q \leq X$, including the diagonal, with denominator $\pi(X)^2$. At every fixed rank k and prime ℓ , the probability that ℓ is the k th smallest distinct prime factor of $|F(p, q)|$ has a limit $D_{F,k}(\ell)$; zero outputs are excluded from the event. The maximal separated reversal count $N_F(k)$ is finite and attained, and

$$\limsup_{k \rightarrow \infty} \frac{\log \log(3 + N_F(k))}{k} = \Lambda(m) > 0.$$

The size limit comes first, with family, rank and prime fixed. Outputs are weighted by their prime-pair representations.

4 The exact comparison and uniform estimates

4.1 First differences

For consecutive primes $p < q$ with interior probabilities,

$$M_r(q) = (1 - \eta_p)M_r(p) + \eta_p M_{r-1}(p).$$

Direct subtraction gives

$$D_{\eta,k}(q) - D_{\eta,k}(p) = \eta_p \eta_q \{M_{k-2}(p) - T_\eta(p, q)M_{k-1}(p)\}, \quad (6)$$

where $T_\eta(p, q) = \eta_q^{-1} - \eta_p^{-1} + 1$. When $M_{k-1}(p) > 0$, an ascent is equivalent to $M_{k-2}(p)/M_{k-1}(p) > T_\eta(p, q)$. The ordinary threshold is $q - p + 1$; for the odd law at $p > 2$ it is

$$q - p + 1 - \frac{q - p}{(p - 2)(q - 2)}.$$

4.2 Finite exceptions and symmetric polynomials

Only finitely many coordinates in (4) can be forced ($\eta_p = 1$) or excluded ($\eta_p = 0$). Let f count the forced coordinates. Past them, each contributes a factor z to the generating polynomial; excluded coordinates contribute 1. Retain all interior coordinates, including the exceptional ones, and set

$$w_p = \frac{\eta_p}{1 - \eta_p}, \quad S(p) = \sum_{\substack{q < p \\ 0 < \eta_q < 1}} w_q.$$

For $k > f$, put $r = k - 1 - f$. The common positive product of $1 - \eta_q$ cancels in the ratio, leaving $E_{r-1}(p)/E_r(p)$, where E_r is the elementary symmetric polynomial of degree r in the finite list of interior odds.

Lemma 4.1. *For at least $r \geq 1$ positive weights, let W_{r-1} be the largest sum of $r - 1$ distinct weights. Then*

$$\frac{r}{S} \leq \frac{E_{r-1}}{E_r} \leq \frac{r}{S - W_{r-1}}, \quad (7)$$

and the right denominator is positive.

Proof. The terms of SE_{r-1} which choose a new index total rE_r . The remaining terms are

$$\sum_{|A|=r-1} \left(\sum_{j \in A} w_j \right) \prod_{j \in A} w_j,$$

between zero and $W_{r-1}E_{r-1}$. Rearrangement proves the bounds. At least one positive weight remains after any $r - 1$ are removed. $\square$

This specializes to Wang–Crapis [4, Lemmas 3.1–3.2]; the present use also retains the exact forced-rank shift.

Expanding (4) gives $w_p = \nu/p + O(p^{-2})$. The reciprocal-prime estimate and the convergent quadratic tail give

$$S(p) = \nu\mathcal{L}(p-1) + O_\eta(1), \quad W_{r-1} = O_\eta(1 + \log(r+1)). \quad (8)$$

For the second estimate, dominate all but finitely many odds by C/p and use $p_j \geq j + 2$. The finitely many exceptional odds contribute a fixed constant; their order need not be monotone.

Lemma 4.2 (Uniform sign margins). *Fix $0 < u \leq v$. Uniformly in the band $uk \leq \mathcal{L}(p-1) \leq vk$, as $k \rightarrow \infty$,*

$$\frac{E_{r-1}(p)}{E_r(p)} = \frac{k}{\nu\mathcal{L}(p-1)} + o(1), \quad r = k - 1 - f.$$

Gaps at most H give strict ascents throughout the band if $(H + \nu)v < 1$. Gaps at least D give strict descents there if $Du > 2$, for sufficiently large rank and lower prime.

Proof. The prime number theorem ensures more than r interior primes below p uniformly in this band. Equations (7) and (8) have numerator $k + O_\eta(1)$ and denominator $\nu\mathcal{L}(p-1) + O_\eta(\log(k+1))$, bounded below by a positive multiple of k . This proves the uniform ratio statement.

Writing $g = q - p$, the threshold is $g/\nu + 1 + \rho(q) - \rho(p)$. Its error is bounded by twice the supremum of $|\rho|$ beyond p , even for large g . For $g \leq H$ the limiting lower ratio is $1/(\nu v) > (H + \nu)/\nu$. For $g \geq D$ the upper ratio is $1/(\nu u) + o(1) < D/\nu$. Both margins are strict. $\square$

Without an upper band endpoint, the same estimates give a one-sided bound: on $\mathcal{L}(p-1) \geq bk$ the ratio is at most $1/(\nu b) + o(1)$, uniformly for fixed $b > 0$.

4.3 Every fixed rank

For fixed $r \geq 1$, the ratio tends to zero by (7)–(8). Consecutive sufficiently large primes have gap at least two, so their threshold is uniformly positive. Equation (6) gives strict descent for supported degrees. If $k \leq f$, the tail is zero. If $k = f + 1$, the remaining degree is zero, its predecessor coefficient vanishes, and the same identity again gives descent. Thus every fixed-rank tail is nonincreasing.

Once all ascent indices lie below an integer I , a separated family has size at most I . Attainable sizes form a nonempty bounded subset of $\mathbb{N}$ (zero is attainable), so their maximum is attained. The canonical laws have no forced coordinates and have positive supported tails, giving strict eventual decrease.

5 From short gaps to separated reversals

The following finite argument supplies the counting step and its loss.

Lemma 5.1 (Periodic composite barriers). *Fix integers $D \geq H + 1$ and $M = D!$. Let $\mathcal{S}$ be a nonempty finite set of integer starts such that*

$$X \leq n, \quad n + H \leq Y, \quad \#([n, n + H] \cap \{\text{primes}\}) \geq 2, \quad X \geq M + 2.$$

There are m separated pairs of consecutive-prime gaps, a gap at least D followed by one at most H , all endpoints in $[X, Y]$, with

$$|\mathcal{S}| \leq 2(M + H)(m + 1). \tag{9}$$

Proof. Choose consecutive primes $p < q$ in each window, and assign them to $c = \lfloor (p - 2)/M \rfloor$. Since $cM + j$ is composite for $2 \leq j \leq D$, and $c \geq 1$,

$$cM + D + 1 \leq p < q \leq (c + 1)M + 1.$$

Indeed a prime q beyond the last endpoint but within $H < D$ of p would lie in the next composite block. A short pair cannot cross that barrier.

Starts assigned to cell c lie between $cM + D + 1 - H$ and $(c + 1)M + 1$, so there are at most $M + H$ of them. Order the occupied cells $c_0 < \dots < c_{t-1}$ and choose one pair in each, with lower-prime index u_j . Between chosen

primes in two successive occupied cells, a consecutive-prime gap crosses the destination cell's whole composite block. Its length is at least D , and it precedes the chosen short gap there.

Take destination cells $c_2, c_4, \dots, c_{2m}$, where $m = \lfloor (t-1)/2 \rfloor$. Choose each crossing after the lower prime selected in the preceding occupied cell. The unused occupied cell between successive destinations ensures $a_i + 1 < d_{i+1}$. Every crossing lies between selected endpoints, so remains in $[X, Y]$. Finally $t \leq 2(m+1)$ and $|\mathcal{S}| \leq (M+H)t$, proving (9). $\square$

No uniform distribution among cells is needed. Empty stretches are harmless, since a crossing still exists between selected occupied cells.

Fix $u, v > 0$ with $(H+\nu)v < 1$ and choose an integer $D \geq H+1$ with $D > 2/u$. If all primes from X through $Y+H$ lie in the sign band, the two lemmas imply, at sufficiently large k ,

$$G_H(Y) \leq X + B(N_\eta(k) + 1), \quad B = 2(D! + H). \quad (10)$$

At most X lower primes are deleted below X ; each retained short gap supplies a window starting at its lower prime. The upper endpoint is $Y+H$, since a counted gap may end above Y . All constants are fixed before k grows.

Corollary 5.2 (Positive-power transfer). *Suppose that for fixed H and $\delta > 0$, for every sufficiently large X at least X^δ integer starts in $[X, 2X]$ have two primes in their width- H windows. For every $0 < a < 1/(H+\nu)$,*

$$N_\eta(k) \geq \mathcal{E}(ak) \quad \text{for all sufficiently large } k.$$

Proof. Choose $a < b < 1/(H+\nu)$ and $X = \lceil \mathcal{E}(bk) \rceil$. Every endpoint in $[X, 2X+H]$ has $\mathcal{L}(p-1) = bk + o(1)$. Fixed $u < b < v < 1/(H+\nu)$ therefore provide the sign band. The cell lemma gives $N_\eta(k) + 1 \geq X^\delta/B$. Since $\log(X^\delta/B) = \delta \exp(bk) + O(1)$ and $a < b$, the right side eventually exceeds $1 + \mathcal{E}(ak)$. $\square$

We use the existing quantitative input

$$\#\{n \in [X, 2X] \cap \mathbb{N} : \#([n, n+600] \cap \{\text{primes}\}) \geq 2\} \geq X^{1/3} \quad (11)$$

for all sufficiently large X . Its maintained declaration is `PrimeGaps.eventually_many_prime_windows_600`. Section 11 records its provenance. Quantitative prime-cluster theory, including Maynard's work [5], is the analytic background. We use this count, not merely infinitude of bounded gaps. With $\nu = 1$ the corollary applies at $a = 1/602 < 1/601$.

6 Proof of the exact spectrum

6.1 Lower bound: arbitrary frequency subsequences

Fix H and $0 < a < \gamma_H/(H + \nu)$. Choose

$$0 < g < \gamma_H, \quad 0 < v_0 < v < \frac{1}{H + \nu}, \quad a < gv_0,$$

then $0 < u < a$ and $a < b < gv_0$. There are arbitrarily large integers Y with $3 + G_H(Y) > \mathcal{E}(g\mathcal{L}(Y))$. Assign $k = \lceil \mathcal{L}(Y)/v_0 \rceil$. The bounds $v_0k - v_0 < \mathcal{L}(Y) \leq v_0k$ imply $3 + G_H(Y) > \mathcal{E}(bk)$ at arbitrarily large assigned ranks.

Take $X = \lceil \mathcal{E}(uk) \rceil + 1$. The primes between X and $Y + H$ lie in a band with positive lower coefficient and upper coefficient v . Equation (10) applies. If $N_\eta(k) < \mathcal{E}(ak)$, then

$$3 + G_H(Y) \leq 3 + X + B(1 + \mathcal{E}(ak)) < \mathcal{E}(bk)$$

eventually, a contradiction. Therefore $N_\eta(k) \geq \mathcal{E}(ak)$ for arbitrarily large k . Supremizing over H proves the lower bound $\Lambda(\nu)$ in (5). The argument keeps the actual frequency subsequence; it does not replace it by an all-large- X hypothesis.

6.2 Upper bound: each gap retains its own cutoff

For fixed integer $g \geq 2$ and $\varepsilon > 0$, the one-sided ratio bound implies that an ascent across a gap of size g must satisfy, apart from fixed initial exceptions,

$$\mathcal{L}(p - 1) < \left(\frac{1}{g + \nu} + \varepsilon \right) k. \quad (12)$$

The same estimate excludes all gaps greater than J above $\mathcal{L}(p - 1) = uk$ if $1/(J + 1 + \nu) < u$. In both statements, the reciprocal error is uniformly small beyond the lower prime.

Consequently

$$N_\eta(k) \leq T_k + \sum_{2 \leq g \leq J} G_g(U_g(k)), \quad (13)$$

where we may take

$$T_k = \lceil \mathcal{E}(uk) \rceil + O(1), \quad U_g(k) = \left\lceil \mathcal{E} \left(\left(\frac{1}{g + \nu} + \varepsilon \right) k \right) \right\rceil + 1.$$

Each witness has a distinct ascent index. There are at most T_k initial ones; all others have an actual gap $g \leq J$ and are counted in that gap's term. Overcounting by the cumulative G_g only strengthens the upper bound.

Let $A > \Lambda(\nu)$. Choose $0 < u < A$ and then J as above. For each fixed g and $\delta > 0$, eventually

$$\log \log(3 + G_g(U)) \leq (\gamma_g + \delta)\mathcal{L}(U).$$

Since $\gamma_g/(g + \nu) \leq \Lambda(\nu)$, choose $\varepsilon, \delta > 0$ small enough that

$$(\gamma_g + \delta) \left(\frac{1}{g + \nu} + \varepsilon \right) < A \quad (2 \leq g \leq J).$$

There are only finitely many such comparisons. Rounding changes their double-logarithmic cutoffs by $o(1)$. Each term of (13), including T_k , has rate strictly below A . A fixed finite sum has rate at most the greatest term rate. Letting A decrease to $\Lambda(\nu)$ proves the matching bound.

The input (11) implies $\gamma_{600} = 1$. A given consecutive pair belongs to at most 601 integer-start windows of width 600, so the associated cumulative gap count has a positive-power lower bound along large cutoffs. The trivial bound $G_H(X) \leq X$ gives the reverse inequality on its exponent. Hence $\Lambda(\nu) \geq 1/(600 + \nu) > 0$. Together with the fixed-rank proof this completes Theorem 3.2.

For the canonical upper bound use the minimum eventual gap, two, and $\nu = 1$ in (12). Reducing ε slightly absorbs the rounding, leaving all ascents below $\mathcal{E}((1/3 + \varepsilon)k)$. Their number bounds $N_\eta(k)$. Together with Corollary 5.2, this proves Theorem 3.3.

7 A finite phase profile

Theorem 7.1. *Let H_* be the least $H \geq 2$ for which $\gamma_H = 1$. This integer exists, and for every fixed $\nu > 0$,*

$$\Lambda(\nu) = \max_{2 \leq H \leq H_*} \frac{\gamma_H}{H + \nu}.$$

For all sufficiently large ν ,

$$\Lambda(\nu) = \frac{1}{H_* + \nu}, \quad \frac{1}{\Lambda(\nu)} - \nu = H_*. \quad (14)$$

Proof. Existence follows from $\gamma_{600} = 1$. If $H > H_*$, then $\gamma_H/(H + \nu) \leq 1/(H + \nu) < 1/(H_* + \nu)$. For each $H < H_*$ put $\gamma = \gamma_H < 1$. With

positive denominators, the inequality $\gamma/(H + \nu) < 1/(H_* + \nu)$ is equivalent to $(1 - \gamma)\nu > \gamma H_* - H$. It holds beyond a finite threshold. Take the greatest threshold over the finitely many smaller classes. $\square$

Here full frequency means only $\gamma_H = 1$. It need not imply positive relative density or a fixed power lower bound at every large cutoff. The theorem identifies a finite maximum without determining its entries numerically.

8 Arithmetic realization by affine prime sums

We prove Theorem 3.4 without discarding exceptional primes.

8.1 Fixed-modulus limits and independence

For fixed Q , the prime number theorem in arithmetic progressions gives

$$\frac{\#\{p \leq X : p \text{ prime}, p \equiv a \pmod{Q}\}}{\pi(X)} \longrightarrow \frac{1}{\varphi(Q)} \quad ((a, Q) = 1).$$

Nonunit prime classes contain only primes dividing Q , so their normalized contribution vanishes. Full Cartesian sampling of the ordered pairs makes each fixed pair of unit classes have limiting probability $\varphi(Q)^{-2}$.

Take Q to be the product of any fixed finite set of primes. The Chinese remainder theorem identifies the unit-pair space with the product of the local unit-pair spaces. Divisibility of F at a prime depends only on the local pair, so the limiting divisibility indicators are independent, with exact law

$$\eta_\ell = \frac{\#\{(u, v) \in (\mathbb{F}_\ell^\times)^2 : \prod_j (c_j(u + v) + d_j) = 0\}}{(\ell - 1)^2}. \quad (15)$$

For fixed k, ℓ the rank event uses only the primes at most ℓ . Summing their joint atoms gives (1).

If $F(p, q) = 0$, then some $p + q = -d_j/c_j \leq |d_j| \leq \sum_j |d_j|$. Only finitely many such pairs exist, independently of X . Their normalized contribution is zero. The diagonal was included throughout; its $\pi(X)$ pairs are also negligible against $\pi(X)^2$. No modulus or rank was allowed to grow with X .

8.2 Exact local counts

Set $T_\ell = \{t \in \mathbb{F}_\ell : \prod_j (c_j t + d_j) = 0\}$. There are $\ell - 1$ unit pairs with sum zero, and $\ell - 2$ with any specified nonzero sum. Thus at every prime,

$$\eta_\ell = \frac{(\ell - 2)|T_\ell| + \mathbf{1}_{0 \in T_\ell}}{(\ell - 1)^2}. \quad (16)$$

If a factor is identically zero modulo ℓ , $T_\ell = \mathbb{F}_\ell$ and $\eta_\ell = 1$. An empty root set gives zero, and coincident roots are counted once. The formula covers all coefficient, shift, determinant, and small-prime exceptions.

Outside the finite set dividing the nonzero coefficients, nonzero shifts, or determinants $c_i d_j - c_j d_i$, there are m distinct roots. At most one shift is zero; let z_0 be one if it is and zero otherwise. Then

$$\eta_\ell = \frac{m(\ell - 2) + z_0}{(\ell - 1)^2}, \quad \eta_\ell^{-1} = \frac{\ell}{m} - \frac{z_0}{m^2} + O_F(\ell^{-1}). \quad (17)$$

The law satisfies (4) with $\nu = m$. Theorem 3.2 proves the arithmetic theorem.

The simplest instances illustrate the exceptional coordinate:

$F(p, q)$	η_2	$\eta_\ell \quad (\ell > 2)$
$p + q$	1	$1/(\ell - 1)$
$p + q + 1$	0	$(\ell - 2)/(\ell - 1)^2$
$2(p + q) - 1$	0	$(\ell - 2)/(\ell - 1)^2$.

The forced factor 2 in $p + q$ shifts subsequent ranks by one. Both odd examples have exactly the canonical odd law. All three have rate $\Lambda(1)$, despite different finite-rank behavior. Equations (5) and (14) also show how sufficiently large fixed numbers of distinct affine factors recover H_* . There is no uniformity claim when the family varies with the rank.

9 Late ascents and bounded prime gaps

For a canonical law, a *late ascent at coefficient b* is an ascent across consecutive primes $p < q$ with $\mathcal{L}(p - 1) \geq bk$. Unbounded supply requires existence beyond every prescribed lower bound on both k and p .

Theorem 9.1 (Recurrence equivalence). *Let $H \geq 1$ be an integer and $1/(H + 2) < b < 1/(H + 1)$. Gaps at most H recur arbitrarily far out if and only if for every K, Q there exist $k \geq K$ and consecutive primes $p < q$, with $p \geq Q$ and $\mathcal{L}(p - 1) \geq bk$, at which both canonical densities ascend.*

Proof. Choose $b < a < 1/(H + 1)$. To a sufficiently late gap at most H assign $k = \lfloor \mathcal{L}(p - 1)/a \rfloor$. Then $k \rightarrow \infty$ with p , $\mathcal{L}(p - 1) \geq ak \geq bk$, and $\mathcal{L}(p - 1)/k \rightarrow a$. A small fixed band around a in Lemma 4.2 gives both ascents.

Conversely, $b > 1/(H + 2)$ and the one-sided ratio bound exclude gaps at least $H + 1$ from sufficiently late ascents. In the odd law the correction to the ordinary threshold has absolute value at most $1/(p - 2)$, absorbed by the fixed strict margin. Unbounded lower primes therefore give recurring gaps at most H . $\square$

Corollary 9.2 (Critical location). *Suppose gaps at most $h \geq 1$ recur arbitrarily far out and every sufficiently late consecutive gap is at least h . The critical canonical late-ascent coefficient is $1/(h + 1)$: every smaller positive coefficient has unbounded simultaneous ascent supply; every larger coefficient forces both densities to descend for all sufficiently large ranks and lower primes in that region. No assertion at equality is made.*

Proof. Use the forward rank assignment on gaps at most h and the one-sided upper ratio bound on gaps at least h . $\square$

For every fixed $b > 1/5$, an ascent in either canonical law at sufficiently large k, p with $\mathcal{L}(p - 1) \geq bk$ must cross a gap strictly below four. Gaps between odd primes are positive even integers, so it must be two. An independently proved unbounded supply would therefore imply infinitely many twin primes. That supply is not established here. A large finite ascent certifies a finite prime pair, not recurrence.

10 Implications and scope

The spectrum gives a quantitative dictionary. Bounded-gap counts yield reversal counts through the finite selection lemma; the upper comparison shows that no other late ascents can improve the rate. Finite changes in an admissible law preserve that rate. Many different affine prime-pair expressions consequently share a spectrum despite different forced primes and small ranks.

Location and frequency encode different information. Corollary 9.2 depends on the least recurring gap. The rate $\Lambda(\nu)$ depends on how frequently bounded gap classes occur. Infinitude alone need not give $\gamma_H > 0$: a hypothetical count growing as a power of $\log X$ has $\gamma_H = 0$. This is why bare infinitude cannot replace the input to Corollary 5.2.

For Goldbach-type expressions, the arithmetic theorem describes factorization in the full representation-weighted ensemble. It gives no lower bound for representations of each fixed even integer. Fixed-modulus independence likewise does not supply uniform estimates when a prime cutoff grows with input size. Moving-tail distribution laws require additional uniformity.

Repeated rises coexist with a final decreasing tail at each rank. This is a statement about a probability sequence’s shape, rather than cancellation in a prime-counting error term. No implication for the Riemann hypothesis, Robin’s inequality, or colossally abundant numbers is proved. Such an application would require a new quantitative bridge to the corresponding error term or optimization. The completed result is the spectrum and its precise prime-gap transfers. Current exploratory approaches to independent ascent supply are outside this paper.

11 Formalization and reproducibility

The principal maintained owners are:

- `Assembly/ReciprocalSmoothSpectrum.lean`: all-rank finite maxima and Theorem 3.2;
- `Assembly/Headline.lean`: Theorem 3.3;
- `Assembly/AffineSpectrum.lean`: Theorem 3.4 and arithmetic phase recovery;
- `Assembly/GapSpectrumPhase.lean`: Theorem 7.1;
- `Assembly/GapRecurrence.lean` and `Proof/Upper/GapScaleCutoff.lean`: Theorem 9.1, Corollary 9.2, and the conditional inverse twin-prime transfer.

Paths are relative to `PrimeFactorOscillations/`. The theorem ledger maps intermediate arguments to declarations. Ordinary prose sometimes combines several compiled lemmas; these combinations are distinguished from individually named declarations.

The analytic inputs are the reciprocal-prime estimate, fixed-modulus prime progression limits and (11). Narrow leaves of the pinned Erdős 690 development supply ordinary density and symmetric bounds; its conditional classification is not an input. The width-600 count is extracted from a separately pinned prime-cluster development. This paper proves the selection, transfers, spectrum and arithmetic realization; it does not reproduce the underlying analytic prime-distribution proofs.

The public Challenge states two headline theorems over Mathlib primitives. Solution contains their actual proofs. The two deliberate Challenge placeholders are absent from the Solution closures. The recorded audit checks the public compiled types and twelve theorem closures, finding only

`propext`, `Classical.choice`, and `Quot.sound`. An explicit hypothesis in a conditional theorem, such as `ascent supply`, remains a hypothesis; an axiom audit does not remove it.

The dependency record and build guide specify Windows Lean 4.34.0, source revisions and artifact hashes. The project reuses the Erdős dependency’s published cache without rebuilding or redistributing it. Additional analytic compatibility artifacts still require an existing matching view; a complete fresh-machine bootstrap is not claimed. The verification record separates initial proof checks from this editorial revision. Branch links provide mutable navigation; the records retain checked source hashes.

No numerical experiment proves an asymptotic step here. The exact inequality $1/602 < 1/601$ supplies the canonical constant’s strict margin; no effective onset rank is given. Official Comparator/NanoDa replay and independent expert review have not been claimed. Kernel checking verifies proof terms for the stated definitions under the reported foundations. Source fidelity, attribution, novelty and significance require scrutiny beyond the kernel. AI assistance was used for research, proof development and exposition.

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